Observation of Gravitational Waves from a Binary Black Hole Merger

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ABSTRACT

On September 14, 2015 at 09:50:45 UTC the two detectors of the Laser Interferometer Gravitational-wave Observatory simultaneously observed a transient gravitational-wave signal. The signal sweeps upwards in frequency from 35 to 250 Hz with a peak gravitational-wave strain of 1.0e-21. It matches the waveform predicted by general relativity for the inspiral and merger of a pair of black holes and the ringdown of the resulting single black hole. The signal was observed with a matched-filter signal-to-noise ratio of 24 and a false alarm rate estimated to be less than 1 event per 203 000 years, which corresponds to a significance greater than 5.1 sigma.

I. INTRODUCTION

The first direct detection of gravitational waves has long been regarded as one of the most important goals of modern physics. General relativity predicts that a system of two orbiting compact objects loses energy through the emission of gravitational radiation, causing the orbit to decay. For binary black holes this inspiral ends in a merger.

II. DETECTORS

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The LIGO detectors are Michelson interferometers with 4 km long arms. Each detector employs a power-recycled Fabry-Perot Michelson configuration. The strain sensitivity of the instrument is limited by seismic noise at low frequency, thermal noise in the mirror coatings at intermediate frequency, and photon shot noise at high frequency.

III. SEARCH METHOD

We use matched filtering against a bank of template waveforms. For a given template, the matched-filter signal-to-noise ratio is computed from the two detector data streams, coincidence is required within the 10 ms light travel time between the sites, and a chi-squared consistency test is applied to reject non-Gaussian transients.

IV. RESULTS

The reconstructed source parameters place the component masses at 36 and 29 solar masses and the final black hole mass at 62 solar masses, with 3 solar masses radiated as gravitational waves. The luminosity distance is estimated at 410 Mpc.

V. DISCUSSION

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The detection is consistent with the general-relativistic prediction for a binary black hole coalescence. No evidence for deviations from general relativity was found in the accessible parameter space. The rate of such mergers is estimated to be between 2 and 600 Gpc^-3 yr^-1 at 90% confidence.

VI. CONCLUSION

This observation confirms the existence of binary stellar-mass black hole systems and opens a new era of gravitational-wave astronomy. Future observing runs with improved sensitivity will increase the detection rate substantially.

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